DO NOW — RECALL

Lesson 056 · Tree Traversals: Post-order and Breadth-fi

Name: _____ Class: _____ Date: _____

1. What is a leaf node?

2. What is BFS's underlying structure?

3. What order does postorder use?

4. A binary tree has root 10, children 5 and 15, and node 5 has right child 7 only. Trace its postorder traversal, using left subtree, right subtree, root.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Tree node with no children.

2. Model answer

A queue.

3. Model answer

Left subtree, right subtree, root.

4. Model answer

7, 5, 15, 10. Finish the left subtree, then the right subtree, then visit the root.

By the end, you can…

• Post-order visits both child subtrees before its parent. Traverse left subtree, right subtree, then root.

• Breadth-first visits by levels, left-to-right with a queue that stores the next vertices.

• Traversal result depends on the precise tree and stated child order. BFS queue changes must be shown to justify an answer.

Vocabulary — know these terms

postorder — Left-right-root tree traversal.

BFS — Breadth-first level exploration.

queue — FIFO worklist.

recursive — Process invoking itself on children.

NEW LEARNING

Read, annotate and explain

Understand

Post-order visits both child subtrees before its parent. Traverse left subtree, right subtree, then root.

Model

Breadth-first visits by levels, left-to-right with a queue that stores the next vertices.

Evaluate

Traversal result depends on the precise tree and stated child order. BFS queue changes must be shown to justify an answer.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Give postorder for tree root5, left2, right7; node2 has left1.

Explain

PostOrder(5) waits for PostOrder(2), which prints 1 then 2. The right call prints 7; only then is 5 printed: 1,2,7,5. null returns without printing. Node supplies Key, Left and Right.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Root8, children3 and10, with 3 children1 and6, 10 right child14. Give postorder. [6]

Question 2

Give breadth-first order for same tree, and first two queue changes. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why postorder is useful for deleting tree structures but BFS for shortest unweighted-edge distance. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Breadth-first tree traversal uses a stack to process deepest nodes first.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.